

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Architected a **distributed executor** exposing typed **REST APIs** over Docker and serverless compute with an **S3** store at **sub-200ms p99**, secured with **OAuth/RBAC**.
- Led a production **multi-agent platform** on **AWS** handling **millions of daily transactions** at **99%+ uptime**, and led its architecture and design reviews.
- Worked backwards from **customer requirements** with product and cross-functional teams, testing prototypes with users and shipping **integrations** to production.
- Shipped **SSE-based streaming** agent delivery with **guardrails** and real-time **observability**, cutting redundant API calls **90%**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by moving LLM serving to an **async, autoscaled** architecture.
- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale feature and dataset preparation feeding training and **evals**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime** on **AWS**.
- Cut production escalations **60%** in one quarter by standardizing API contracts and failure handling, driving **root cause analysis** and post-incident reviews.
- Led **architecture and code reviews** and mentored engineers on system design, reliability, and secure API practices.
- Cut nightly batch aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** on a financial RDBMS with indexed views and partitioning.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- **Raft**-consensus key-value store with **linearizable reads** and **log compaction**, plus a companion VPC control plane simulator.

LangFetch

tinyurl.com/LangFetch

- **LangGraph** agent that decomposes natural-language **financial queries** into dependency-ordered **SQL** subqueries with **Pinecone** retrieval, served via **FastAPI** as an **MCP** server.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Java, Python, TypeScript, SQL, C++

Distributed Systems: Microservices, REST APIs, Event-Driven Architecture, SQS, Kafka, High Availability, Scalability

AWS: Lambda, S3, SQS, RDS, DynamoDB, EC2, API Gateway

Practices: Design Reviews, Code Reviews, Mentoring, CI/CD, Agile, Testing, Operational Excellence

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer